

02 – HTML

Writing the content of web pages

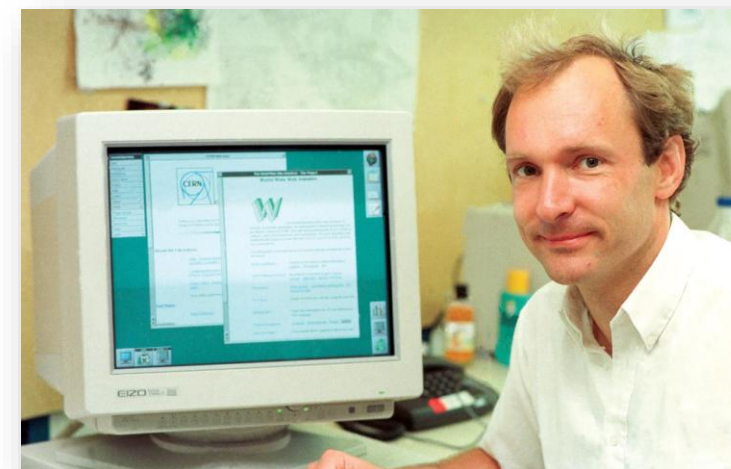
Jérôme Landré – jerome.landre@ju.se

OUTLINE

- 1) Introduction to HTML
- 2) History
- 3) HTML structure and syntax
- 4) HTML tags
- 5) HTML attributes
- 6) Examples
- 7) Best practices
- 8) Resources about HTML

1) INTRODUCTION TO HTML

- **Hypertext Markup Language (HTML)** is the standard markup language for documents designed to be displayed in a web browser. It defines the **content** and **structure** of web content. It is often assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript. (HTML: 2025)
- HTML was invented by Tim Berners-Lee at CERN: Geneva: Switzerland.
- The first version had only 18 tags and was created to share information with hypertext links between researchers at CERN. (Berners-Lee: 1990)



1) INTRODUCTION TO HTML

- The purpose of HTML is the **separation of content** and **presentation**: improved **semantics**: **multimedia** support: and **cross-platform** compatibility.
- HTML is a **language** to **describe** the **content** of web pages using **tags**.
- There are **opening tags** and **closing tags**:

<tag> ... **</tag>**

```
<> index.html ×
project-2025 > <> index.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Web Dev is Fun</title>
7  </head>
8  <body>
9      <h1>Web Development Fundamentals</h1>
10     <p>Please remember, Web Dev is Fun! :)</p>
11 </body>
12 </html>
13
```

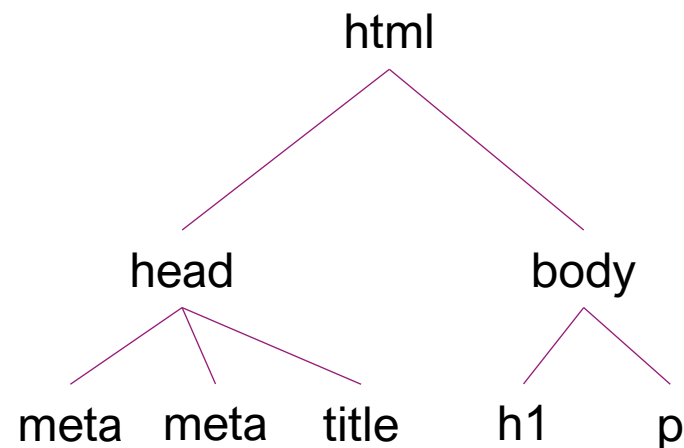
1) INTRODUCTION TO HTML

- Remark (very important): an HTML document is a **tree**!

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2) HISTORY

- **A long time ago...**
 - HTML (1990): Basic structure and links.
 - HTML 2.0 (1995): Forms: form elements.
 - HTML 3.2 (1997): Tables: applets: text flow around images.
 - HTML 4.01 (1999): CSS support: scripting: frames.
 - XHTML (2000): Stricter syntax: XML-based.
 - HTML5 (2014): Major revision: new APIs: semantic elements: multimedia support. (HTML: 2025)
- **Why HTML5?**
 - Need for better multimedia support without plugins (Flash: Silverlight).
 - Improved semantics for accessibility and SEO.
 - Mobile and cross-device compatibility.
 - Standardization by W3C (W3C: 2025) and WHATWG.



3) HTML STRUCTURE AND SYNTAX

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13
```

- <!DOCTYPE html>: HTML5 document
- <html>: the content starts...
- <head>: **invisible information** to give details on how the file is encoded: the link to stylesheets: javascript... </head>
- <body>: **visible information** that will appear on the displayed page in the browser. </body>
- </html>: ...end of content

3) HTML STRUCTURE AND SYNTAX

- Be careful on the order of the elements (tags): we want to obtain a document which is a tree!
- Tags have to be nested such that elements are all completely within each other: without overlapping:

~~`<p>This is <em>very <strong>wrong</em>!</strong></p>`~~

`<p>This <em>is <strong>correct</strong>.</em></p>`

```

- DOCTYPE: html
- html lang="en"
  - head
    - #text: ␣
    - title
      - #text: Sample page
    - #text: ␣
  - #text: ␣
  - body
    - #text: ␣
    - h1
      - #text: Sample page
    - #text: ␣
    - p
      - #text: This is a
      - a href="demo.html"
        - #text: simple
      - #text: sample.
    - #text: ␣
    - #comment: this is a comment
    - #text: ␣␣

```

```

<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Sample page</title>
  </head>
  <body>
    <h1>Sample page</h1>
    <p>This is a <a href="demo.html">simple</a> sample.</p>
    <!-- this is a comment -->
  </body>
</html>

```

- HTML user agents (e.g.: web browsers) then parse this markup: turning it into a DOM (Document Object Model) tree. A DOM tree is an in-memory representation of a document.
- DOM trees contain several kinds of nodes: in particular a DocumentType node: Element nodes: Text nodes: Comment nodes: and in some cases ProcessingInstruction nodes.

4) HTML TAGS

- Tags can be grouped by specific **semantic** family
- In HTML: "**semantic**" refers to the use of tags that **clearly describe the meaning** and **purpose** of the content they enclose: making the **structure more understandable** to both browsers and humans.
- Semantic HTML Explained
 - Semantic HTML elements: like <header>: <nav>: <main>: <article>: and <footer>: provide context for what type of content they contain—such as navigation links: main content: or supplementary material. Unlike generic tags like <div> or : semantic tags add meaning: not just layout or style.
- Benefits of Semantic HTML
 - Improved accessibility: Screen readers and assistive technologies can navigate and interpret webpages better.
 - SEO advantages: Search engines can identify and rank content more accurately.
 - Better readability and maintenance: Developers and collaborators can easily understand the intent and structure of the page.
 - Semantic HTML ensures that content is not only well-presented but also meaningful to machines and people alike.

<!DOCTYPE>: Declares the document type and HTML version.

<html>: Root element of an HTML page.

<head>: Container for metadata (title: charset: links to CSS/JS).

<title>: Defines the title of the document (shown in browser tab).

<body>: Contains the visible page content.

<header>: Represents introductory content: typically a group of introductory aids.

<nav>: Defines a set of navigation links.

<main>: Specifies the main content of a document.

<section>: Defines a section in a document.

<article>: Represents self-contained content.

<aside>: Defines content aside from the content (like a sidebar).

<footer>: Defines a footer for a document or section.

- <h1>-<h6>: Headings: from most important (<h1>) to least (<h6>).
- <p>: Defines a paragraph.
- <a>: Defines a hyperlink.
- : Embeds an image.
- : Defines an unordered list.
- : Defines an ordered list.
- : Defines a list item.
- <div>: Defines a division or section in a document.
- : Defines a section in a document (inline).
- <video>: Embeds video content.
- <audio>: Embeds sound content.
- <canvas>: Used to draw graphics: on the fly: via scripting (usually JavaScript).
- <form>: Defines an HTML form for user input.
- <input>: Defines an input control.
- <button>: Defines a clickable button.

5) HTML ATTRIBUTES

- Elements can have **attributes**: which **control** how the elements work. In the example below: there is a hyperlink: formed using the **a** element and its **href attribute**:

```
<a href="demo.html">simple</a>
```

5) HTML ATTRIBUTES

- **Attributes** are placed **inside the start tag**: and consist of a **name** and a **value**: separated by an "=" character. The attribute value can remain unquoted if it doesn't contain ASCII whitespace or any of " ' ` = < or >. Otherwise: **it has to be quoted using either single or double quotes**. The value: along with the "=" character: can be omitted altogether if the value is the empty string.

```
<!-- empty attributes -->
<input name=address disabled>
<input name=address disabled="">

<!-- attributes with a value -->
<input name=address maxlength=200>
<input name=address maxlength='200'>
<input name=address maxlength="200">
```

To **avoid errors**: I suggest to **always** use **" everywhere**:

```
<input name="address" maxlength="200" disabled="">
```

5) HTML ATTRIBUTES

- In HTML: several attributes are **MANDATORY** for several tags:

4.8.3 The **img** element

An **img** element represents an image.

At least one of the **src** and **srcset** attributes must be present.

Content attributes:

Global attributes

alt — Replacement text for use when images are not available

src — Address of the resource

srcset — Images to use in different situations, e.g., high-resolution displays, small monitors, etc.

sizes — Image sizes for different page layouts

crossorigin — How the element handles crossorigin requests

usemap — Name of [image map](#) to use

ismap — Whether the image is a server-side image map

```



<div id=very-large></div> <!-- Everything after this div is below the viewport -->


```

ed by the element

this image for presentation

ed by the element

6) EXAMPLES

<> index-00.html X

html-css > <> index-00.html > ...

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>My first web page</title>
7  </head>
8  <body>
9    <h1>Welcome!</h1>
10   <p>
11     I am happy to see you on this page.
12   </p>
13 </body>
14 </html>
15
```

Welcome!

I am happy to see you on this page.

<> index-01.html X

html-css > <> index-01.html > ...

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>My first web page</title>
7  </head>
8  <body>
9      <!-- SEMANTIC LAYOUT -->
10     <header>Welcome to my web page...</header>
11     <nav>Home   Teaching   Research   About</nav>
12     <main>
13         <article>Teaching Web Development Fundamentals</article>
14         <article>Teaching Databases</article>
15         <article>Research in Image Processing</article>
16
17         <aside>A new project will start soon.</aside>
18     </main>
19     <footer>&copy; JL - 2025</footer>
20 </body>
21 </html>
22
```

Welcome to my web page...
Home Teaching Research About
Teaching Web Development Fundamentals
Teaching Databases
Research in Image Processing
A new project will start soon.
© JL - 2025

CSS is needed here!

<> index-02.html X

html-css > <> index-02.html > ...

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>My first web page</title>
7  </head>
8  <body>
9    <!-- EMBEDDING A VIDEO -->
10   <video controls width="90%">
11     <source src="http://commondatastorage.googleapis.com/gtv-videos-bucket/sample/BigBuckBunny.mp4" type="video/mp4">
12     Your browser does not support the video tag.
13   </video>
14 </body>
15 </html>
16 |
```



<> index-03.html X

html-css > <> index-03.html > ...

```
1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>My first web page</title>
7    </head>
8    <body>
9      <!-- SIMPLE FORM -->
10     <form action="/submit" method="post">
11       <label for="name">Name:</label>
12       <input type="text" id="name" name="name" required placeholder="Please type in your name...">
13       <input type="submit" value="Submit">
14     </form>
15   </body>
16 </html>
17
```

Name:

7) BEST PRACTICES

- All **tags** and **attributes** must be written in **lower case**
- All attributes must get a value using this format: `attributename="value of attribute"`
- Use **semantic** elements for better accessibility and SEO.
- Always include **alt** text for images.
- Validate your HTML using the **W3C Validator** (<https://validator.w3.org/>).
- Keep structure and presentation separate (use CSS for styling).

8) RESOURCES ABOUT HTML

- MDN Documentation [MDN Web Docs: HTML5](#)
- W3schools online tutorials [W3Schools HTML5 Tutorial](#)
- HTML5 official documentation [HTML5 Specification \(W3C\)](#)

REMEMBER:

WEB DEVELOPMENT FUNDAMENTALS

WEB DEV IS FUN

ANY QUESTIONS?